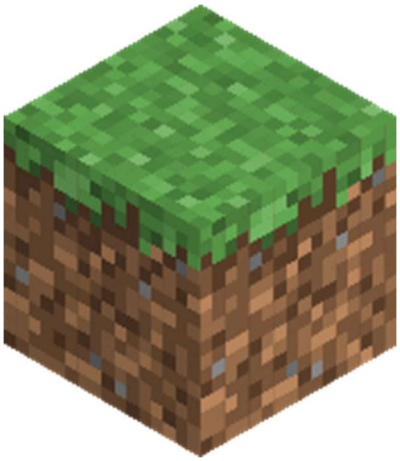
How to build a Minecraft Server



By TomMustBe12

A complete guide to building and setting up your very own Minecraft Java Edition server, with mods, plugins, and optimizations!

What you can expect:

Basic Setup

- Learn how to download and install the server!
- Join and give yourself admin commands.

Allowing others to join

- Learn methods of port forwarding or tunneling.
- Set up whitelist and bans.

Customization

- Install plugins and mods to your server.
- Learn how to find good plugins or mods.

https://tommustbe12.com/blog/how-to-build-a-minecraft-server/

Getting the requirements

For players, Java is built into Minecraft, but for server owners, you'll have to download the Developer Kit. Pay attention here, because it is a little bit hard to get the right Java version.

Depending on the version of the server you are going to make, you will need a different JDK. The JDK is the basis of the server's code, and without it, your server will crash fatally.

Version	JDK
1.21+	JDK 21
1.18-1.20.6	JDK 17
1.17.x	JDK 16
1.8-1.16.5	JDK 8
1.7.x and below	JDK 7

JDK 16 and lower are rarely used as of today since newer versions exist. Plus, there are issues regarding direct downloading because Oracle has licensing on some of them.

Server version & type

Now that you have your JDK set up, you can start researching what server version and type you would like for your server.

Now, you may be wondering what a server 'type' is. A server type is a type of software your server runs on. You might have heard of Paper or Spigot before. These are server types!

I prefer Paper. This is because it balances quality and performance almost perfectly. It also uses your CPU cores more efficiently, meaning higher TPS (server's frames) and less lag.

Downloads:

Paper: https://papermc.io/downloads/paper

Java: https://minecraft.net/en-us/download/server

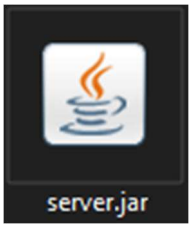
Fabric (mods): https://fabricmc.net/use/server/

Other types include Forge, Spigot, and tweaked versions of these types.

Installation of the server

With your .jar file ready to go, your server can be started! It's simple to start and will be one of the easiest parts of this server setup.

First, take your 'server.jar' file you downloaded (names vary) from your type's website. You need to create a new folder on your home or Desktop. Drag the .jar into your empty folder.



Now, let's create a Windows batch file. This will allow you to double click or run the batch file and start your server quickly. So, create a new text file, and rename it to 'start.bat'. Make sure view file name extensions is turned on in File Explorer.

Now, in the start.bat file, put this text:

```
java -Xms2G -Xmx4G -jar server.jar nogui
```

Keep in mind that the -Xms and -Xmx values are the minimum and maximum RAM amounts. You need to change these depending on how much RAM your server has!

Also, change 'server.jar' to whatever your .jar file's name is. Then, save your file, and run it!

Running the server & EULA

Now that you have run your batch file to start the server, you should see a black box with a bunch of text! Now let's see what to do next.

Creating your server from here is easy! However, there is one important detail before your server fully runs. You have to accept Minecraft EULA (basically Minecraft's way of making sure servers are safe for players).

Now, find the eula.txt file in the folder you made earlier. Open it, and change 'eula=false' to 'eula=true'. This means you take full responsibility for your server and will be held accountable if you do anything against the EULA.

After changing that, close out of your current server batch file, and re-run 'start.bat'. Now, your server should spin to life! When it is done creating the world and the server, it will say 'Done'. You'll notice it starts creating a bunch of files in the folder. These are for the world chunks and player inventories.

Joining and administration

Now that you have your server set up, it's time to join it and find out how to give yourself and other players admin commands, allowing for more customization and moderation on your server.

Minecraft servers are super easy to join! Go to Multiplayer in Minecraft and click "Add Server". Then, in the 'Server Address' area, put 'localhost'. This will direct Minecraft to join the server you are hosting right now!

Now that you are in your server, go to your black command box (the one that said "Done" when you started your server) and type 'op username'. This will make whoever's username typed an operator, meaning you can use admin commands like /ban, /kick, and more!

In the future, you can run /op in-game instead of the command box. However, if you ever lose access to your server or can't log in, that black box can run any command you'd like!

Allowing others to play

Sure, playing by yourself is a whole adventure, but playing with others is the reason you created the server in the first place!

This part of the server process can be really challenging, especially depending on how fast your internet is, if you can port forward, or something else. However, I'll show you the best methods.

Playit.gg is one of the easiest and simplest ways to make your server public. Plus, it is relatively fast. It is 100% free, so you don't have to pay anything. You can download it at https://playit.gg. Create a Java Edition tunnel, and they will give you an address players can connect to.

Port forwarding is letting others directly connect to your network. Go to 192.168.1.1 or 10.0.0.1 in your browser, and you are likely to find a router page. Log in and navigate to security or network > port forwarding. To allow connections to your Minecraft server, add '25565' as TCP/UDP port, and share your IP address with players of your server.

Warning: giving others your IP could be dangerous. Make sure you trust them before giving them this.

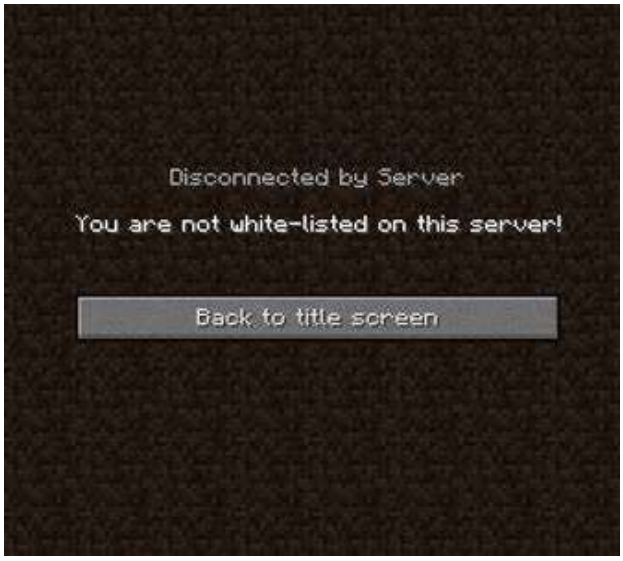
Whitelists, kicks, & bans

Whitelists allow you to allow only certain players such as you and your friends. Kicks mean temporarily disconnecting the player, and bans mean making the player never able to join again.

Whitelists are necessary nowadays. If you have a server, it is prone to griefing (especially playit.gg and other tunneling services). If you want your world to not be destroyed, whitelists are very recommended.

To enable whitelists, use /whitelist on. This turns on the whitelist meaning only players on the whitelist can join.

Adding players to the whitelist is just as easy. Use /whitelist add username to add any username to the whitelist. You'll have to do this every time you want another player on the server.



This is the message players get when they are not whitelisted. They won't be able to join or connect. In the black command box (the log) you will see that they disconnect immediately after joining.

Whitelists, kicks, & bans

Kicking and banning players that break rules on your server is necessary for server functionality and player experience.

Kicking a player is very easy to do. All you have to do is run /kick player in game, the 'player' being their username. Once you do that, they will be kicked. However, kicking still lets them join again.

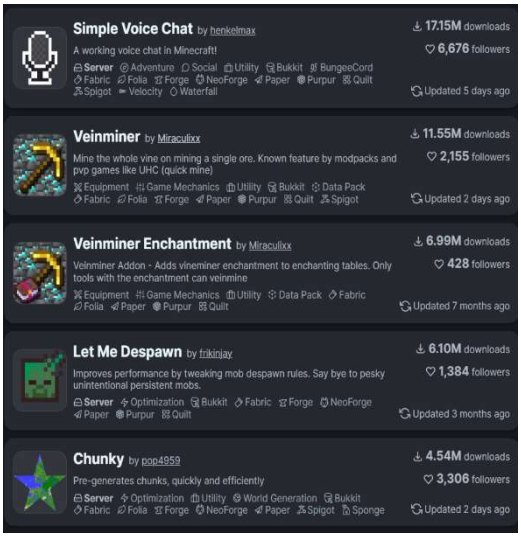
Banning, however, is permanent, and banning a player means they can't join again until you unban them. You can ban a player with /ban player reason (again, 'player' being their player username) and the reason will tell them why they are banned every time they try to join again.

Change your mind? You can unban a player with /pardon player. There is no command called /unban, so lots of server operators that are new to Minecraft servers will not know how to unban. The correct command for this is /pardon.

Customization

Now that you have your server set up and ready for your friends to join, you might want to have a better experience than just normal Minecraft.

Many people prefer using plugins or mods – these make the game a bit more interesting and fun for players. Here's how they work and how you can add them!



If you installed Paper or Spigot earlier when setting up your server, you're able to use plugins! Plugins are bits of code that run on the server's side. This means that everyone that connects to your server can join normally, but you can still add minigames, custom commands, and more to your server.

To install a plugin, you can go to spigot's website (spigotmc.org) or modrinth's website (modrinth.com) and search their many plugins available. To install, just drag them into the 'plugins' folder in your server!

Customization

Forge or Fabric servers can support mods – mods require the server and the players to have them, but they add a lot more flexibility and possibilities.

Didn't install Paper or Spigot? That's fine – if you installed Forge or a Fabric server, you can install mods! Mods can be found on modrinth.com as well. Mods allow more customization, like custom mobs, textures, structures, and even dimensions!

First, download the mod you'd like from your preferred mod provider. Download the correct version depending on if you have Fabric, or Forge. Then, drop the .jar file into the 'mods' folder. Make sure you have that mod installed on your own computer too, all players will have to have it installed for everyone to have the experience!

If you don't have the necessary mods installed, you won't be able to join your server. Make sure everybody on the server has the mod installed locally.

Finishing up

You're finally done setting up your server! Have fun with your friends, or set up many plugins/mods to make a public fun server!

Great job setting up your server! Now, you can share the server address with your friends or community! If you port forwarded/tunneled your server correctly, you should start seeing players join!

You can contact me on Discord (@tommustbe12) or on my website (https://tommustbe12.com/contact/) if you have an issue with this or something needs to be changed.

Thanks for using this tutorial, have fun with your new Minecraft server!

Sincerely, TomMustBe12